

Guide to model assessment

This is a brief guide to what things to do/consider when assessing model fit. Let's use the Study Habits data without the outliers.

```
Study_Habits<-read.csv("Stats_study_habits_noout.csv",header=T)
head(Study_Habits,2)
summary(Study_Habits)
```

Note that many of the binary/categorical predictors are coded as numbers. This means that you need to `as.factor` them. Cohort is either 2012 or 2013 and is an indicator of whether the student that responded to the questionnaire on study skills took the course in 2012 or 2013. Gender is "M" and "F". A.level is the grade in Maths/Stats on a scale from 0 to 6 and is considered to be a continuous variable.

```
grade.all.lm<-lm(Grade~as.factor(Cohort)+as.factor(Interesting)+as.factor(Enjoy)
+as.factor(Social.Net)+Gender+A.Level+Study.Skills,data=Study_Habits)
display(grade.all.lm)
```

```
## lm(formula = Grade ~ as.factor(Cohort) + as.factor(Interesting) +
##      as.factor(Enjoy) + as.factor(Social.Net) + Gender + A.Level +
##      Study.Skills, data = Study_Habits)
##               coef.est coef.se
## (Intercept)      20.61    20.08
## as.factor(Cohort)2013   4.33     3.05
## as.factor(Interesting)1 10.57     3.59
## as.factor(Enjoy)1       2.38     3.58
## as.factor(Social.Net)2   2.48     5.16
## as.factor(Social.Net)3  -0.05     4.89
## GenderM              -4.03     3.20
## A.Level              0.95     3.24
## Study.Skills          1.29     0.33
## ---
## n = 85, k = 9
## residual sd = 13.86, R-Squared = 0.39
```

What to do with the output of `display()`

Q1: For each predictor in the model:

1. Interpret the coefficients associated with it
2. Check the signs of the coefficients make sense
3. Determine whether the predictor is significant at the 5% level

Q2: Is the model a good fit? Consider the following:

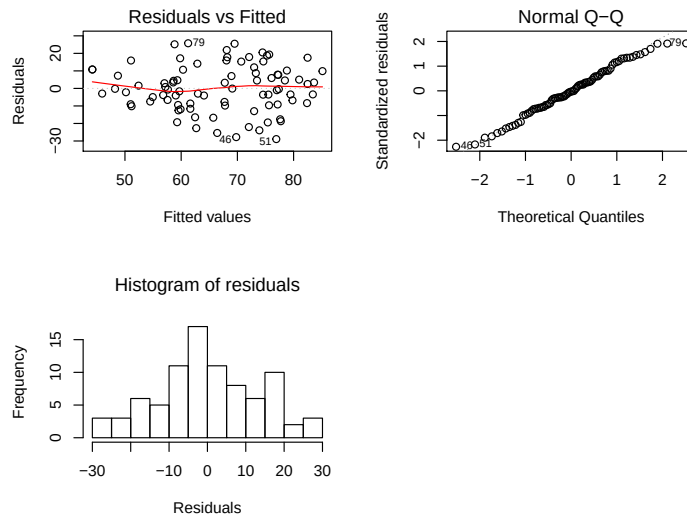
1. R^2
2. The residual sd (check the absolute range of `Grade` using `abs(range(Study_Habits$Grade))`)
3. The significance of the coefficients. If there are too many non-significant coefficients it makes sense to get rid of some especially if the sign does not makes sense. Which 2 would you get rid of and why?

(If you want to see the Adjusted R^2 or the F-statistic, look at `summary()`)

It's not all about the diagnostics! You need to look at residual plots too!

The same residual plots are relevant for multiple linear regression as for simple linear regression.

```
par(mfrow=c(2,2))
plot(grade.all.lm,which=c(1,2))
hist(grade.all.lm$residuals,main="Histogram of residuals",font.main=1,xlab="Residuals")
```



Q3: Based on the plots can we conclude that the residual assumptions of constant variance and normality of residuals hold? Look for the following:

1. Does the fitted vs residual plot look like random scatter (no trends, curvature, funnel shape)?
2. Do the points in the Q-Q plot sit approximately on the diagonal (no bow/s shapes)?
3. Does the histogram of the standardised residuals look approximately normal (no bimodality, skewness, kurtosis)?

Q4: If some of the predictors are non-significant you can consider removing them – I’m not a big fan of doing this especially if the non-significance is borderline (e.g. a p-value of 0.06) and when the sign of the predictor has the right sign/size. If you do remove them, does anything change?

Q5: Should you add interactions? What happens if you do?